

TLQLDMR: A Transfer Learning Quantile Regression Model for Wind Power Forecasting under Extreme Weather Conditions

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ABSTRACT

Accurate wind power forecasting under extreme weather conditions is critical for grid stability and renewable energy integration, yet remains a formidable challenge due to severe data scarcity (extreme events constitute less than 5% of historical records) and substantial distribution shifts between normal and extreme meteorological regimes. While kernel-based methods, particularly Support Vector Regression (SVR) and Large-margin Distribution Machine-based Regression (LDMR), have demonstrated strong performance in various regression tasks due to their theoretical guarantees and robustness on limited data, their application to extreme weather forecasting faces critical limitations: the i.i.d. assumption underlying standard kernel methods is fundamentally violated under cross-domain scenarios, and the symmetric variance minimization in LDMR implicitly encourages symmetric error distributions, which conflicts with the asymmetric nature of quantile-based interval prediction required for uncertainty quantification. To extend kernel-based regression to extreme weather forecasting, we propose Transfer Learning Quantile Large-margin Distribution Machine for Regression (TLQLDMR), which addresses these challenges through three synergistic innovations: (i) an asymmetric variance regularizer that preserves LDMR's distributional robustness while accommodating quantile-specific error asymmetry, (ii) a supervised transfer learning framework integrating Maximum Mean Discrepancy (MMD) for feature alignment and weighted pinball loss for label adaptation, enabling effective knowledge transfer from abundant normal-weather data to scarce extreme-weather samples, and (iii) a scalable Nyström-SGD algorithm reducing complexity from $O(N^3)$ to $O(T \cdot N \cdot m)$. Experiments on six wind farms demonstrate that TLQLDMR achieves competitive performance against recent kernel-based methods, ensemble approaches, and deep learning baselines under extreme weather conditions. On the most challenging target farm, TLQLDMR attains RMSE = 7.476 and $R^2 = 0.9553$ for point prediction, and PICP = 0.959 with CWC = 0.268 for 95% interval prediction, demonstrating superior reliability-sharpness trade-off. The method maintains robust performance under measurement noise and exhibits practical scalability (1.67 seconds for $N = 2,000$), making it suitable for real-time SCADA applications where interpretability and theoretical guarantees are prioritized.

1. Introduction

The global transition toward renewable energy has positioned wind power as a cornerstone of sustainable electricity generation, with installed capacity projected to exceed 1,000 GW by 2030 (Wang et al., 2019). However, the inherent intermittency and volatility of wind resources pose significant challenges to power grid stability and dispatch optimization. Accurate wind power forecasting is therefore critical for balancing supply and demand, reducing operational costs, and ensuring reliable grid integration (Wang et al., 2019; Zhang & Wang, 2024). While substantial progress has been achieved in short-term forecasting under normal meteorological conditions, prediction performance can degrade by 30-50% during extreme weather events—such as typhoons, cold waves, or heat waves—when wind power fluctuations are most severe and grid stability is most vulnerable (State Grid Renewable Energy, 2022).

Extreme weather events present a dual challenge for data-driven forecasting models. First, such events are inherently rare, resulting in severe data scarcity: extreme samples typically constitute less than 5% of historical records (State Grid Renewable Energy, 2022). Second, the meteorological dynamics during extreme conditions differ fundamentally from normal regimes, inducing substantial distribution shifts in both feature space (e.g., wind speed, temperature) and label space (power output). For instance, during typhoon events, wind speeds may exceed 25 m/s, triggering turbine cut-out

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operations that produce near-zero power output—a pattern rarely observed in normal weather training data. This creates a covariate shift in the input space and a label shift in the output space, both of which challenge conventional machine learning models that assume training and testing data are independently and identically distributed (i.i.d.) (Mustaffa & Sulaiman, 2025; Chen & Shi, 2025).

Among the diverse landscape of forecasting methodologies, kernel-based approaches—particularly Support Vector Regression (SVR) and its variants—have been extensively applied to wind power forecasting and have demonstrated competitive performance under normal weather conditions (Wang et al., 2019). The SVR methodology continues to evolve, with recent advances including metaheuristic optimization for hyperparameter tuning (Houssein et al., 2025), functional iterative solvers for improved convergence (Meena et al., 2025), and feature selection techniques for enhanced prediction accuracy (Duan et al., 2025). These kernel methods offer several distinctive advantages over black-box deep learning models: strong generalization guarantees rooted in statistical learning theory, effective handling of high-dimensional feature spaces through the kernel trick, convex optimization formulations that guarantee global optima, and robust performance on small-to-medium-sized datasets where deep models tend to overfit. Moreover, their interpretability and theoretical foundations make them particularly attractive for safety-critical applications such as grid dispatch, where model transparency is often required by regulatory standards.

However, existing kernel-based wind power forecasting studies predominantly focus on normal weather conditions with deterministic point prediction objectives. Traditional SVR minimizes the ϵ -insensitive loss to estimate the conditional mean, achieving robust point predictions under Gaussian noise assumptions. To address the need for uncertainty quantification, Support Vector Quantile Regression (SVQR) extends SVR by replacing the symmetric ϵ -loss with the asymmetric pinball loss, enabling estimation of conditional quantiles for prediction intervals (Ye et al., 2025b, 2023). Recent SVQR variants incorporate twin-hyperplane designs (Ye et al., 2023), sparse kernel approximations (Wang & Xu, 2024), and nonlinear feature selection (Ye et al., 2025b) to improve scalability and robustness. Despite these methodological advances, the application of kernel-based methods to extreme weather forecasting—where cross-domain distribution shift and probabilistic interval prediction are simultaneously required—remains largely unexplored, presenting both challenges and opportunities for methodological extension.

Despite these advances, both SVR and SVQR share a fundamental limitation: they optimize only the minimum margin (the distance from the decision boundary to the closest support vectors) while ignoring the statistical distribution of margins across the entire dataset. This focus on worst-case samples can lead to overfitting and poor generalization, particularly when training data contains outliers or label noise—a common occurrence in SCADA systems during extreme weather. To overcome this limitation, the Large-margin Distribution Machine-based Regression (LDMR) paradigm was introduced, which explicitly optimizes both the mean and variance of the margin distribution (Zhang & Zhou, 2019; Gupta & Gupta, 2021). By concentrating prediction errors tightly around zero, LDMR achieves superior generalization compared to classical SVR on a wide range of regression tasks (Gupta & Gupta, 2021; Hazarika & Gupta, 2023). Variants of LDMR have been successfully applied to wind turbine fault detection (Tang et al., 2019), imbalanced classification (Cheng et al., 2016a,b), and structural health monitoring (Yang et al., 0), demonstrating its robustness in noisy industrial environments. However, LDMR has not been systematically applied to wind power forecasting, particularly under extreme weather conditions where both distributional robustness and probabilistic interval prediction are required. These characteristics—combined with LDMR's theoretical foundations—make it a promising candidate for extension to extreme weather forecasting scenarios.

However, extending kernel-based methods from normal weather forecasting to extreme weather scenarios introduces three critical challenges that require methodological innovation. First, the i.i.d. assumption—that training and testing data are drawn from the same distribution—is a foundational premise of both SVR/SVQR and LDMR. While this assumption holds reasonably well within a single weather regime, it is systematically violated when transferring knowledge from normal (source domain) to extreme (target domain) conditions. For example, in our preliminary analysis of six wind farms, the Wasserstein distance between normal and extreme power distributions ranges from 5.96 to 21.06, indicating substantial distributional discrepancy. A model trained solely on abundant normal weather data will inevitably exhibit performance degradation when deployed under extreme conditions, regardless of how well the margin distribution is optimized within the source domain. While transfer learning and domain adaptation techniques have been extensively studied in computer vision (Tao et al., 2012) and sensor drift compensation (Yan et al., 2022; Guo et al., 2022; Zhai et al., 2023; Se et al., 2023), their principled integration with kernel-based regression for wind power forecasting remains an open research question.

Second, there exists a fundamental theoretical tension between LDMR's distributional robustness mechanism and the requirements of quantile regression. The core principle of LDMR—minimizing the variance of prediction residuals—implicitly encourages a symmetric, concentrated error distribution around zero. This is theoretically appropriate and empirically effective for conditional mean estimation, as evidenced by LDMR's superior performance over SVR in numerous regression benchmarks. However, quantile regression at level $\tau \neq 0.5$ requires an inherently asymmetric residual distribution: for example, when estimating the 95th percentile (upper prediction bound), approximately 95% of errors should be negative (under-predictions) and only 5% positive (over-predictions). Naively applying LDMR's variance minimization to quantile regression will force the error distribution toward symmetry, thereby degrading interval calibration and violating the target coverage probability. This theoretical conflict has received limited attention in the literature, as existing LDMR studies focus primarily on mean regression, and existing SVQR studies do not incorporate distributional robustness. Reconciling these two objectives—distributional robustness and quantile-specific asymmetry—requires a principled modification of the LDMR framework.

Third, existing transfer learning approaches for wind power forecasting predominantly adopt unsupervised domain adaptation strategies, focusing exclusively on aligning feature distributions (e.g., via Maximum Mean Discrepancy) while treating target domain labels—even when available in small quantities—as entirely unused supervision signals (Tao et al., 2012; Guo et al., 2022). This design choice is reasonable when target labels are completely unavailable, but is suboptimal for extreme weather forecasting, where scarce target samples (typically 3-5% of total data) represent high-value information that captures critical domain-specific dynamics such as cut-out behavior and power curtailment. A more effective transfer learning framework should leverage both feature-space alignment (to reduce covariate shift) and label-space supervision (to correct prediction bias), strategically weighting target samples to guide model adaptation. This supervised transfer learning paradigm has been successfully applied in computer vision and natural language processing, but its integration with kernel-based quantile regression for wind power forecasting remains unexplored.

To address these challenges and extend kernel-based regression to extreme weather forecasting, we propose the Transfer Learning Quantile Large-margin Distribution Machine for Regression (TLQLDMR), a principled extension of the classical LDMR paradigm specifically designed for cross-domain probabilistic wind power prediction. Our approach preserves the theoretical strengths of kernel-based methods—convex optimization, generalization guarantees, and robustness on limited data—while introducing three synergistic innovations tailored to the extreme weather scenario.

First, we introduce a novel asymmetric variance regularization mechanism that reconciles the tension between distributional robustness and quantile estimation. Unlike standard LDMR, which implicitly encourages symmetric error distributions through variance minimization, our formulation employs asymmetric variance regularization based on algebraic (signed) residuals, thereby compressing overall error dispersion while preserving the requisite skewness for accurate quantile prediction. This innovation enables TLQLDMR to achieve both superior point prediction accuracy (through tight error concentration) and well-calibrated prediction intervals (through correct error asymmetry)—a capability that neither standard LDMR nor SVQR alone can provide. This reconciles the tension between distributional robustness (tight margin concentration) and quantile estimation (asymmetric error distribution), enabling TLQLDMR to achieve both superior point prediction accuracy and well-calibrated prediction intervals.

Second, we develop a supervised transfer learning framework that integrates two complementary adaptation mechanisms. In the feature space, we employ Maximum Mean Discrepancy (MMD) minimization to align the source (normal weather) and target (extreme weather) distributions in the Reproducing Kernel Hilbert Space (RKHS), reducing covariate shift and ensuring that the learned feature representation is domain-invariant (Tao et al., 2012). Crucially, in the label space, we introduce a weighted pinball loss mechanism that assigns substantially higher weights ($C_T/C_S \approx 3000$) to scarce target domain samples, treating them as high-value corrective anchors that guide the model toward extreme weather dynamics. This dual-space adaptation strategy enables TLQLDMR to leverage abundant source data for structural learning while prioritizing target data for domain-specific refinement, addressing the label neglect problem in existing unsupervised domain adaptation methods. This is particularly important for extreme weather forecasting, where every extreme sample carries valuable information about rare but critical operating conditions such as cut-out events and power curtailment.

Third, recognizing that the standard quadratic programming (QP) formulation of kernel-based transfer learning incurs $O(N^3)$ computational complexity due to the need to compute and invert the full kernel matrix, we propose a scalable optimization algorithm that combines Nyström approximation—which constructs an explicit low-rank feature mapping using $m \ll N$ landmark points—with stochastic gradient descent (SGD), reducing complexity to $O(T \cdot N \cdot m)$.

This enables practical deployment on large-scale SCADA datasets that accumulate tens of thousands of samples annually.

We evaluate the proposed method on a publicly available multi-site wind farm SCADA dataset (State Grid Renewable Energy, 2022), encompassing six wind farms with capacities ranging from 36 MW to 200 MW. Extreme-weather events are defined by statistical thresholds (98th-percentile wind speed, 2nd/98th-percentile temperature) and a physical cut-out condition (wind speed above 25 m/s). We compare TLQLDMR against three categories of baselines adapted to the wind power forecasting task: (i) recent kernel-based methods including optimization-enhanced SVR (HHO-SVR) (Houssein et al., 2025), functional iterative SVR (FLSVR) (Meena et al., 2025), associated road analysis SVR (ARA-SVR) (Duan et al., 2025), twin support vector quantile regression (TSVQR) (Ye et al., 2023), sparse support vector quantile regression (SSVQR) (Wang & Xu, 2024), and nonlinear feature selection SVQR (NFS-SVQR) (Ye et al., 2025b); (ii) ensemble methods including KMeans-GBT (Rizkallah, 2025), Random Forest (RF-WPF) (Mustaffa & Sulaiman, 2025), Broad Random Forest (BRF-WPF) (Chen & Shi, 2025), and Quantile Gradient Boosted Regression (QGBR) (Hu & Liu, 2025); and (iii) deep learning approaches including hybrid deep learning interval prediction (HybridDL-Interval) (Mao et al., 2025), attention-based multi-step quantile RNN (AMQRNN) (Ye et al., 2025a), and conformalized quantile regression MLP (CQR-MLP) (Hu et al., 2025). On the most shifted farm (Farm 6, exhibiting the largest domain shift with Wasserstein distance = 25.12), TLQLDMR attains the best deterministic performance ($\text{RMSE} = 7.476$, $R^2 = 0.9553$), while in 95% interval prediction it reaches $\text{PICP} = 0.959$ with the lowest CWC (0.268). The improvement remains stable under injected noise for both point and interval metrics, and ablation experiments confirm that transfer learning, variance regularization, and MMD alignment each contribute to final performance. In scalability analysis, Nystrom + SGD completes training at $N = 2,000$ in 1.67 seconds, while full QP routes are not feasible beyond $N = 800$.

It is important to clarify the scope and positioning of this work. TLQLDMR is designed for scenarios where kernel-based methods' distinctive advantages—theoretical guarantees, interpretability, sample efficiency, and convex optimization—are prioritized over the raw predictive power of large-scale deep architectures. Such scenarios are common in practice: (i) extreme weather events are inherently rare, providing limited training data (less than 5% of historical records) where deep models tend to overfit, (ii) grid dispatch applications often require model transparency and explainability for regulatory compliance, and (iii) real-time forecasting systems benefit from the computational efficiency and stability of convex optimization. Our contribution is to extend the LDMR framework with quantile-aware asymmetry and supervised domain adaptation, providing a theoretically grounded and practically deployable option for extreme weather forecasting when these priorities apply. We do not claim that TLQLDMR universally outperforms deep learning methods in all data-rich settings; rather, we demonstrate that kernel-based methods, when properly extended, remain a viable and competitive choice for extreme weather forecasting under the constraints commonly encountered in real-world wind farm operations.

The remainder of this paper is organized as follows. Section 2 reviews the theoretical foundations of SVQR, LDMR, and transfer learning with MMD, establishing the building blocks for our method. Section 3 presents the TLQLDMR formulation, including the asymmetric variance regularizer, supervised transfer framework, and scalable optimization algorithm. Section 4 describes the dataset, experimental setup, and comprehensive evaluation results across deterministic prediction, probabilistic interval forecasting, ablation studies, and computational efficiency analysis. Finally, Section 5 concludes the paper and discusses future research directions.

2. Related Work

This section briefly reviews the theoretical foundations related to the proposed method, including Support Vector Quantile Regression, Large Margin Distribution Machine for Regression, and Transfer Learning with Maximum Mean Discrepancy.

2.1. Support Vector Quantile Regression (SVQR)

Traditional Support Vector Regression (SVR) relies on the ϵ -insensitive loss function, which assumes a symmetric error distribution (e.g., Gaussian) and estimates the conditional mean of the target variable. However, in scenarios like partial shading or wind curtailment, the noise distribution is often asymmetric and heteroscedastic. To address this, Support Vector Quantile Regression (SVQR) was introduced to estimate conditional quantiles, providing a comprehensive view of the outcome distribution. Recent SVQR-family developments include twin-hyperplane designs,

sparse/approximate kernel formulations, and nonlinear feature-selection variants, which improve scalability and robustness in non-Gaussian environments (Ye et al., 2023; Wang & Xu, 2024; Ye et al., 2025b).

The core of SVQR is the non-symmetric pinball loss function, defined as:

$$L_{\tau}(u) = \begin{cases} \tau u, & u \geq 0 \\ (\tau - 1)u, & u < 0, \end{cases} \quad (1)$$

where $\tau \in (0, 1)$ indicates the target quantile. Unlike SVR, SVQR finds a decision boundary that divides the data such that a proportion τ of the samples lie below the boundary. The optimization problem minimizes the structural risk along with the empirical risk measured by the pinball loss:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n L_{\tau}(y_i - (\mathbf{w}^T \phi(\mathbf{x}_i) + b)), \quad (2)$$

this formulation allows SVQR to capture heavy-tailed distributions and outliers more effectively than mean-based regression.

2.2. Large Margin Distribution Machine for Regression (LDMR)

While SVR and SVQR focus on maximizing the minimum margin (the distance from the closest instances to the hyperplane), they largely ignore the statistical distribution of the margin over the entire dataset. The **Large Margin Distribution Machine (LDMR)** was proposed to improve generalization by explicitly optimizing the distribution of the margins (or residuals). From early large-margin distribution formulations to least-squares and mode-seeking variants, LDMR has evolved into a practical framework for robust learning under noisy conditions (Zhang & Zhou, 2014, 2019; Gupta & Gupta, 2021; Hazarika & Gupta, 2023).

The key idea of LDMR is to not only maximize the margin of the support vectors but also to minimize the variance of the error distribution for all samples. A typical LDMR formulation for regression integrates two additional statistical regularization terms into the objective function:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + \lambda_1 \text{Var}(\text{Residuals}) + \lambda_2 \text{Mean}(\text{Residuals}) + C \sum_{i=1}^n L(y_i, f(\mathbf{x}_i)), \quad (3)$$

by constraining the variance, LDMR forces the prediction errors to be clustered tightly around zero, leading to a more compact and robust hypothesis space. However, this variance minimization implicitly encourages symmetric error distributions. Our work extends this concept by introducing an asymmetric variance regularization mechanism suitable for quantile regression, which preserves the distributional robustness of LDMR while accommodating the inherent asymmetry required for accurate quantile estimation.

2.3. Transfer Learning and Maximum Mean Discrepancy (MMD)

Standard machine learning models operate under the assumption that training (source) and testing (target) data are drawn from the same probability distribution. When this assumption is violated—such as in wind power forecasting across different weather regimes—the model performance degrades significantly. Transfer learning aims to transfer knowledge from a label-rich source domain to a target domain with different but related distributions. MMD-based adaptation has been widely used in transfer scenarios because it provides a non-parametric discrepancy measure in RKHS and integrates naturally with kernelized objectives (Tao et al., 2012; Yan et al., 2022; Guo et al., 2022; Zhai et al., 2023; Se et al., 2023).

A fundamental challenge in transfer learning is measuring and reducing the distribution discrepancy. Maximum Mean Discrepancy (MMD) is a widely used non-parametric metric that measures the distance between two distributions in a Reproducing Kernel Hilbert Space (RKHS). Given source data $\{\mathbf{x}_{S,i}\}$ and target data $\{\mathbf{x}_{T,j}\}$, the squared MMD distance is defined as:

$$\text{MMD}^2(D_S, D_T) = \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_{\mathcal{H}}^2, \quad (4)$$

where $\phi(\cdot)$ is the feature mapping induced by the kernel. By minimizing the MMD distance during training, the model learns a feature representation where the source and target domains are statistically aligned, enabling the classifier or

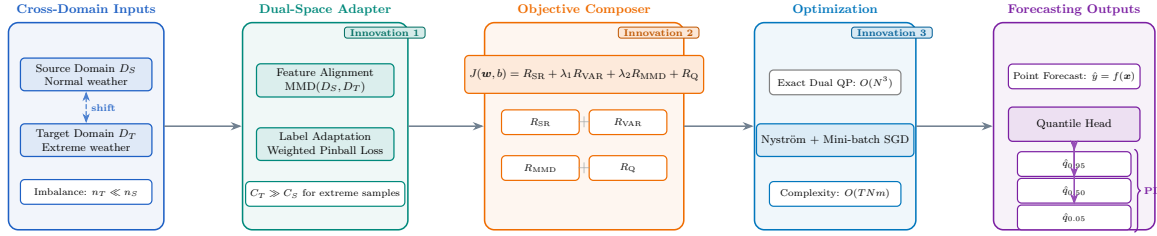


Fig. 1: The overall framework of the proposed TLQLDMR model. Source domain (normal weather) and target domain (extreme weather) data are mapped into a high-dimensional RKHS via kernel mapping. The joint optimization objective integrates structural risk, distribution variance, MMD-based domain adaptation, and weighted pinball loss. The model outputs probabilistic prediction intervals for robust wind power forecasting under extreme conditions.

regressor trained on the source domain to generalize well to the target domain. This alignment is particularly crucial for extreme weather forecasting, as it ensures that the learned predictor is robust to the feature-space shift between normal and extreme meteorological regimes.

3. Proposed Methodology

In this section, we elaborate on the construction of the Transfer Learning Quantile Regression Model (TLQLDMR). We first define the necessary notations, then present the final objective function which integrates Structural Risk, Distribution Variance, Domain Adaptation (MMD), and Weighted Pinball Loss. Finally, we describe the constraints introduced to handle the non-differentiability of the Quantile Loss.

3.1. Notations

Consider the problem of wind power forecasting under extreme weather conditions. We define the **Source Domain (Normal Weather)** as $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, containing abundant conventional weather samples, and the **Target Domain (Extreme Weather)** as $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, containing scarce extreme weather samples. The input space $\mathcal{X} \subseteq \mathbb{R}^d$ represents feature vectors constructed via sliding window processing, and the output space $\mathcal{Y} \subseteq \mathbb{R}$ represents wind power. The sample sizes satisfy $n_T \ll n_S$.

Due to drastic meteorological changes during extreme events, the joint distributions differ significantly: $P_S(\mathbf{X}_S, \mathbf{Y}_S) \neq P_T(\mathbf{X}_T, \mathbf{Y}_T)$. This distribution discrepancy implies that a model trained solely on P_S will fail on P_T . We define a feature mapping $\phi: \mathcal{X} \rightarrow \mathcal{H}$ that maps the input to a high-dimensional Reproducing Kernel Hilbert Space (RKHS). The prediction model is defined as:

$$f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b, \quad (5)$$

where $\mathbf{w}$ is the weight vector and b is the bias. The quantile $\tau \in (0, 1)$ is set to a specific value (e.g., $\tau = 0.95$) for interval prediction (e.g., predicting the upper bound). Unless otherwise noted, $k(\cdot, \cdot)$ denotes the scalar kernel function, $\mathbf{K}$ denotes the Gram matrix, $\phi(\cdot)$ denotes the implicit RKHS mapping used in the exact kernel formulation, and $\psi(\cdot)$ denotes the explicit Nyström feature map used only in the scalable primal solver.

3.2. TLQLDMR Model Formulation

The overall architecture of the proposed TLQLDMR model is illustrated in Fig. 1. The model integrates four key components: structural risk minimization, asymmetric variance regularization, MMD-based domain adaptation, and weighted pinball loss for supervised transfer learning.

3.2.1. Objective Function

Our optimization goal is to find the optimal $\mathbf{w}, b$ and slack variables ξ, ξ^* to minimize the following objective function J . Let $\bar{u}_S = \frac{1}{n_S} \sum_{i=1}^{n_S} (y_{S,i} - f(\mathbf{x}_{S,i}))$ denote the mean residual over the source domain, which is treated as a

scalar constant during optimization:

$$\begin{aligned}
 \min_{\mathbf{w}, b, \xi, \xi^*} \quad J = & \underbrace{\frac{1}{2} \|\mathbf{w}\|^2}_{(1) \text{ Structural Risk}} \\
 & + \underbrace{\lambda_1 \sum_{i=1}^{n_S} ((y_{S,i} - \mathbf{w}^T \phi(\mathbf{x}_{S,i}) - b) - \bar{u}_S)^2}_{(2) \text{ Distribution Variance (LDMR Core)}} \\
 & + \underbrace{\lambda_2 \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_H^2}_{(3) \text{ Domain Adaptation (MMD)}} \\
 & + \underbrace{C_S \sum_{i=1}^{n_S} L_\tau(y_{S,i}, f(\mathbf{x}_{S,i})) + C_T \sum_{j=1}^{n_T} L_\tau(y_{T,j}, f(\mathbf{x}_{T,j}))}_{(4) \text{ Weighted Pinball Loss}}
 \end{aligned} \tag{6}$$

the physical meaning of each term is as follows:

1. **Structural Risk:** This term controls the complexity of the model and prevents overfitting, embodying the maximum margin principle of SVM.
2. **Distribution Variance:** This term is the core of our proposed Quantile LDMR (QLDMR), incorporating a critical modification to the original LDMR variance.
 - **Theoretical Conflict:** The original LDMR minimizes the variance of residuals, which implicitly encourages a symmetric, concentrated error distribution around zero. However, quantile regression at level $\tau \neq 0.5$ requires an asymmetric residual distribution where approximately τ proportion of errors are negative. Applying a symmetric variance penalty would force the residual distribution towards symmetry, degrading the skewness required for accurate quantile estimation.
 - **Asymmetric Modification:** To resolve this, we employ an **Asymmetric Variance** regularization based on the algebraic (signed) residuals. This design allows the model to reduce error dispersion (compactness) without constraining the distribution's direction, thereby preserving the non-symmetric properties essential for the Pinball Loss and improving interval coverage.
3. **Domain Adaptation (MMD):** This term aligns source and target feature distributions in RKHS by minimizing mean-embedding discrepancy. This alignment ensures that the learned predictor is robust to the feature-space shift between normal and extreme weather regimes. MMD is chosen because it is non-parametric, computationally tractable, and directly compatible with kernelized optimization.
4. **Weighted Pinball Loss:** This term performs label-space adaptation under sample imbalance. Unlike purely unsupervised alignment, the objective explicitly includes target-domain labels:
 - **Label-supervised correction:** extreme-weather target labels constrain the model in regions where source-domain trends are no longer reliable (e.g., high wind speed under cut-out operation).
 - **Imbalance-aware weighting:** by setting $C_T \gg C_S$, prediction errors on scarce target samples receive stronger penalties, preventing optimization from being dominated by abundant source samples.

3.2.2. Constraints

To handle the non-differentiability of the Pinball Loss, we introduce slack variables ξ_k, ξ_k^* to transform it into linear constraints. These constraints apply uniformly to all samples k in the combined training set $D = D_S \cup D_T$, while domain-specific weighting is handled through C_S and C_T in the objective function:

$$\begin{cases} y_k - (\mathbf{w}^T \phi(\mathbf{x}_k) + b) \leq \xi_k, & \forall k \in \{1, \dots, n_S + n_T\} \\ (\mathbf{w}^T \phi(\mathbf{x}_k) + b) - y_k \leq \xi_k^*, & \forall k \in \{1, \dots, n_S + n_T\} \\ \xi_k \geq 0, & \xi_k^* \geq 0, \end{cases} \tag{7}$$

in the objective function, the term $L_\tau(y_k, f(\mathbf{x}_k))$ corresponds to $\tau\xi_k + (1 - \tau)\xi_k^*$.

3.3. Model Optimization

In this section, we derive the dual problem of TLQLDMR to enable efficient solving using Quadratic Programming (QP).

3.3.1. Matrix Reformulation of the Primal Problem

To facilitate the derivation, we rewrite the variance term and the MMD term in the objective function [Eq. (6)] using matrix traces or norms. Let $X = [X_S, X_T]$ denote the entire training set with $N = n_S + n_T$ samples. For consistency with standard derivations, we add a coefficient of $1/2$ to the variance and MMD terms. The primal objective can be reformulated as:

$$\min_{\mathbf{w}, b, \xi, \xi^*} \frac{1}{2} \|\mathbf{w}\|^2 + \frac{\lambda_1}{2} \text{Var}_S(\mathbf{w}, b) + \frac{\lambda_2}{2} \text{MMD}^2(\mathbf{w}) + \sum_{k=1}^N C_k L_\tau(y_k, f(\mathbf{x}_k)) \quad (8)$$

where C_k takes value C_S if sample $k \in D_S$ and C_T if $k \in D_T$.

We introduce two auxiliary matrices to express the variance and MMD terms in Eq. (8) as quadratic forms of the weights:

1. **Variance Matrix $\mathbf{S}$:** To represent the source domain variance $\sum_{i \in D_S} (y_{S,i} - f(\mathbf{x}_{S,i}) - \bar{u}_S)^2$, we define a centering matrix $\mathbf{H}_S = \mathbf{I}_{n_S} - \frac{1}{n_S} \mathbf{1}_{n_S} \mathbf{1}_{n_S}^T$. The variance term can be written as $\|\mathbf{H}_S(\mathbf{y}_S - \mathbf{f}_S)\|^2$. To extend this to the full dataset, we define an $N \times N$ matrix $\mathbf{S}$:

$$\mathbf{S} = \begin{bmatrix} \mathbf{H}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (9)$$

2. **MMD Matrix $\mathbf{M}$:** The MMD term represents the distance between means. We define the MMD coefficient matrix $\mathbf{M} \in \mathbb{R}^{N \times N}$ as:

$$M_{ij} = \begin{cases} \frac{1}{n_S^2} & i, j \in D_S \\ \frac{1}{n_T^2} & i, j \in D_T \\ \frac{-1}{n_S n_T} & \text{otherwise} \end{cases} \quad (10)$$

3.3.2. Representer Theorem and Kernelization

According to the Representer Theorem [Schölkopf et al. \(2001\)](#), the optimal weight vector $\mathbf{w}$ lies in the span of the feature mappings of the training samples. Note that the Representer Theorem applies here because all regularization terms (structural risk, variance, and MMD) can be expressed as functions of inner products in RKHS, preserving the kernel structure. Therefore:

$$\mathbf{w} = \sum_{i=1}^N \beta_i \phi(\mathbf{x}_i) = \Phi \boldsymbol{\beta} \quad (11)$$

where $\Phi = [\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_N)]$ is the feature matrix and $\boldsymbol{\beta} = [\beta_1, \dots, \beta_N]^T$ are the coefficients. Substituting Eq. (11) into the prediction function, we get $f(\mathbf{X}) = \boldsymbol{\beta}^T \mathbf{K} + b \mathbf{1}^T$, where $\mathbf{K} = \Phi^T \Phi$ is the $N \times N$ kernel matrix. Consequently, using Eq. (9) and Eq. (10), the structural risk, variance, and MMD terms can be expressed in terms of $\boldsymbol{\beta}$:

- Structural Risk: $\|\mathbf{w}\|^2 = \boldsymbol{\beta}^T \mathbf{K} \boldsymbol{\beta}$.
- Distribution Variance: The quadratic part with respect to $\boldsymbol{\beta}$ is $\boldsymbol{\beta}^T \mathbf{K} \mathbf{S} \mathbf{K} \boldsymbol{\beta}$. Note that the bias b is eliminated by the centering matrix $\mathbf{H}_S$.
- Domain Adaptation (MMD): $\text{MMD}^2(\mathbf{w}) = \boldsymbol{\beta}^T \mathbf{K} \mathbf{M} \mathbf{K} \boldsymbol{\beta}$.

Combining these, we define the total coefficient matrix $\mathbf{\Omega}$:

$$\mathbf{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K} \quad (12)$$

Remark 1. When $\mathbf{K}$ is positive definite (which holds for common kernels such as RBF on distinct samples) and $\lambda_1, \lambda_2 > 0$, the matrix $\mathbf{\Omega}$ is positive definite. This ensures that $\mathbf{\Omega}$ is invertible and the optimization problem is well-posed. The positive definiteness of $\mathbf{\Omega}$ also guarantees the strong convexity of the objective function, which is essential for both the dual QP formulation and the convergence analysis of the scalable SGD algorithm.

By substituting the kernelized terms and Eq. (12) into Eq. (8), the kernelized primal problem becomes (ignoring constant terms w.r.t b in variance):

$$\min_{\beta, b, \xi, \xi^*} \quad \frac{1}{2} \beta^T \mathbf{\Omega} \beta - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \beta + \sum_{k=1}^N C_k (\tau \xi_k + (1 - \tau) \xi_k^*) \quad (13)$$

subject to the constraints from Eq. (7):

$$y_k - (\mathbf{K}_k \beta + b) \leq \xi_k, \quad (\mathbf{K}_k \beta + b) - y_k \leq \xi_k^*, \quad \xi_k, \xi_k^* \geq 0 \quad (14)$$

3.3.3. Lagrangian and Dual Problem

We introduce Lagrange multipliers $\alpha_k, \alpha_k^*, \mu_k, \mu_k^* \geq 0$ for Eq. (14). The Lagrangian function $\mathcal{L}$ is constructed as:

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} \beta^T \mathbf{\Omega} \beta - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \beta + \sum_{k=1}^N C_k (\tau \xi_k + (1 - \tau) \xi_k^*) \\ & - \sum_{k=1}^N \alpha_k (\xi_k - y_k + \mathbf{K}_k \beta + b) - \sum_{k=1}^N \alpha_k^* (\xi_k^* + y_k - \mathbf{K}_k \beta - b) - \sum (\mu_k \xi_k + \mu_k^* \xi_k^*) \end{aligned} \quad (15)$$

By setting the partial derivatives of Eq. (15) with respect to the primal variables (b, ξ, ξ^*, β) to zero, along with the complementary slackness and primal feasibility conditions, we obtain the KKT conditions:

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \implies \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \quad (16)$$

$$\frac{\partial \mathcal{L}}{\partial \xi_k} = 0, \frac{\partial \mathcal{L}}{\partial \xi_k^*} = 0 \implies \alpha_k \in [0, C_k \tau], \quad \alpha_k^* \in [0, C_k (1 - \tau)] \quad (17)$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = 0 \implies \mathbf{\Omega} \beta - \lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{K} (\alpha - \alpha^*) = 0 \quad (18)$$

Let $\gamma = \alpha - \alpha^*$. From Eq. (18), the optimal β is derived as:

$$\beta = \mathbf{\Omega}^{-1} (\mathbf{K} \gamma + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y}) \quad (19)$$

Substituting Eq. (19) back into Eq. (15), and simplifying using Eq. (16) and Eq. (17), we arrive at the **Dual Problem**, which is a standard Quadratic Programming (QP) problem:

$$\begin{aligned} \min_{\alpha, \alpha^*} \quad & \frac{1}{2} (\alpha - \alpha^*)^T \mathbf{Q}_{dual} (\alpha - \alpha^*) + \mathbf{p}_{dual}^T (\alpha - \alpha^*) \\ \text{s.t.} \quad & \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \\ & 0 \leq \alpha_k \leq C_k \tau, \quad \forall k \\ & 0 \leq \alpha_k^* \leq C_k (1 - \tau), \quad \forall k \end{aligned} \quad (20)$$

where the dual matrix $\mathbf{Q}_{dual}$ and vector $\mathbf{p}_{dual}$ are defined as:

$$\mathbf{Q}_{dual} = \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K}, \quad \mathbf{p}_{dual} = \lambda_1 \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{Y} \quad (21)$$

Once the optimal α, α^* are found using a QP solver for Eq. (20), we can recover β using Eq. (19) and construct the final decision function $f(\mathbf{x})$.

Algorithm 1: TLQLDMR Training Algorithm

Input : Source Data $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, Target Data $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, Quantile τ , Kernel parameters, Regularization parameters λ_1, λ_2 , Pinball Loss weights C_S, C_T .

Output: Optimal dual variables α, α^* and bias b .

// Step 1: Data Preparation

- 1 Combine Source and Target data: $\mathbf{X} = [\mathbf{X}_S, \mathbf{X}_T]$, $\mathbf{Y} = [\mathbf{Y}_S; \mathbf{Y}_T]$, $N = n_S + n_T$;
- 2 Set weights vector $\mathbf{C} \in \mathbb{R}^N$ where $C_k = C_S$ for $k \leq n_S$ and C_T otherwise;

// Step 2: Matrix Construction

- 3 Compute Kernel Matrix $\mathbf{K} \in \mathbb{R}^{N \times N}$ using the specified kernel function;
- 4 Construct Variance Matrix $\mathbf{S}$ using Eq. (9);
- 5 Construct MMD Matrix $\mathbf{M}$ using Eq. (10);

// Step 3: Dual Problem Formulation

- 6 Compute coefficient matrix $\mathbf{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K}$ using Eq. (12);
- 7 Invert $\mathbf{\Omega}$ to fit the dual form;
- 8 Compute Dual Matrix $\mathbf{Q}_{dual} = \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K}$ using Eq. (21);
- 9 Compute Dual Vector $\mathbf{p}_{dual} = \lambda_1 \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{Y}$ using Eq. (21);

// Step 4: Solve QP

- 10 Solve the QP problem in Eq. (20) providing $\mathbf{Q}_{dual}$, $\mathbf{p}_{dual}$, and constraints $0 \leq \alpha_k \leq C_k \tau$, $0 \leq \alpha_k^* \leq C_k(1 - \tau)$;
- 11 **Return** Optimal α, α^* ;

// Step 5: Recover Primal Variables

- 12 Compute $\gamma = \alpha - \alpha^*$;
- 13 Recover β using Eq. (19);
- 14 Compute bias b using complementary slackness conditions (averaged over Support Vectors);

3.4. The Algorithm and Complexity Analysis

The complete training process of the proposed TLQLDMR model is summarized in Algorithm 1. The algorithm mainly consists of matrix construction and solving a Quadratic Programming (QP) problem.

The computational complexity of TLQLDMR primarily stems from three parts: matrix multiplication, matrix inversion, and the QP solver.

1. **Matrix Operations:** The construction of $\mathbf{K}$, $\mathbf{S}$, and $\mathbf{M}$ involves $O(N^2d)$ or $O(N^2)$ operations. The calculation of $\mathbf{\Omega}$ involves matrix multiplications of size $N \times N$, leading to a complexity of $O(N^3)$ in the worst case for dense kernel matrices.
2. **Matrix Inversion:** Inverting the $N \times N$ matrix $\mathbf{\Omega}$ requires $O(N^3)$ complexity in the worst case.
3. **QP Solver:** Solving the standard Quadratic Programming problem for the dual variables typically has a complexity ranging from $O(N^{2.3})$ to $O(N^3)$ depending on the solver implementation (e.g., Interior Point Method).

Therefore, the overall time complexity of the proposed algorithm is approximately $O(N^3)$ in the worst case. Given that $N = n_S + n_T$ and $n_T \ll n_S$, the scale is dominated by the source domain sample size. While this complexity is acceptable for moderate-sized datasets, it becomes prohibitive for large-scale wind power forecasting scenarios involving tens of thousands of samples. To overcome this computational bottleneck and enable practical deployment in real-world wind farm applications, we develop a scalable optimization algorithm in the following subsection.

3.5. Scalable Optimization via Nyström Approximation

To address the computational bottleneck of the standard QP-based solver, we propose a scalable optimization algorithm that combines Nyström approximation with stochastic gradient descent. This approach reduces the time complexity from $O(N^3)$ to $O(T \cdot N \cdot m)$, where T denotes the number of iterations and $m \ll N$ is the number of landmark points.

3.5.1. Nyström Feature Mapping

The core idea is to approximate the implicit infinite-dimensional feature mapping $\phi(\mathbf{x})$ with an explicit finite-dimensional representation. Given the training data, we first select m landmark points $Z = \{z_1, \dots, z_m\}$ from the source domain D_S using K-Means clustering to capture the underlying data manifold.

Let $\mathbf{W} \in \mathbb{R}^{m \times m}$ denote the kernel matrix of landmarks, where $W_{ij} = k(z_i, z_j)$. The explicit feature mapping $\psi : \mathcal{X} \rightarrow \mathbb{R}^m$ is then constructed as:

$$\psi(\mathbf{x}) = \mathbf{W}^{-1/2} [k(\mathbf{x}, z_1), \dots, k(\mathbf{x}, z_m)]^T \quad (22)$$

where $\mathbf{W}^{-1/2}$ is computed via eigendecomposition. This mapping satisfies the approximation property $k(\mathbf{x}_i, \mathbf{x}_j) \approx \psi(\mathbf{x}_i)^T \psi(\mathbf{x}_j)$, enabling us to work directly in the primal space.

3.5.2. Primal Formulation with Explicit Features

With the explicit feature mapping, the prediction function becomes $f(\mathbf{x}) = \mathbf{w}^T \psi(\mathbf{x}) + b$, where $\mathbf{w} \in \mathbb{R}^m$. Let $\Psi_S \in \mathbb{R}^{n_S \times m}$ and $\Psi_T \in \mathbb{R}^{n_T \times m}$ denote the mapped feature matrices for source and target domains, respectively. The regularization matrix $\mathbf{R} \in \mathbb{R}^{m \times m}$ incorporating structural risk, variance, and MMD terms is formulated as:

$$\mathbf{R} = \mathbf{I} + \lambda_1 \underbrace{\Psi_S^T \mathbf{H}_S \Psi_S}_{\text{Variance}} + \lambda_2 \underbrace{(\bar{\Psi}_S - \bar{\Psi}_T)(\bar{\Psi}_S - \bar{\Psi}_T)^T}_{\text{MMD}} \quad (23)$$

where $\mathbf{H}_S = \mathbf{I}_{n_S} - \frac{1}{n_S} \mathbf{1}_{n_S} \mathbf{1}_{n_S}^T$ is the centering matrix, and $\bar{\Psi}_S, \bar{\Psi}_T$ are the mean feature vectors of source and target domains.

3.5.3. Stochastic Gradient Descent Optimization

Instead of solving the dual QP problem, we directly optimize the primal objective using mini-batch stochastic gradient descent with the Adam optimizer. For each sample k in a mini-batch $\mathcal{B}$, the sub-gradient of the Pinball Loss with respect to the prediction is:

$$g_k = \frac{\partial L_\tau}{\partial \hat{y}_k} = \begin{cases} \tau - 1, & \text{if } y_k < \hat{y}_k \\ \tau, & \text{if } y_k \geq \hat{y}_k \end{cases} \quad (24)$$

The weighted sub-gradient is computed as $\tilde{g}_k = C_k \cdot g_k$, where $C_k \in \{C_S, C_T\}$ depends on the domain membership. The parameter gradients are then:

$$\nabla_{\mathbf{w}} J = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \tilde{g}_k \psi(\mathbf{x}_k) + \mathbf{R} \mathbf{w}, \quad \nabla_b J = \frac{1}{|\mathcal{B}|} \sum_{k \in \mathcal{B}} \tilde{g}_k \quad (25)$$

The complete training procedure is summarized in Algorithm 2.

3.5.4. Complexity Analysis of the Scalable Algorithm

The proposed scalable algorithm achieves significant computational savings:

1. **Landmark Selection:** K-Means clustering requires $O(N \cdot m \cdot I_{km})$ operations, where I_{km} is the number of K-Means iterations.
2. **Feature Mapping:** Computing $\mathbf{W}^{-1/2}$ costs $O(m^3)$, and mapping all N samples requires $O(N \cdot m \cdot d)$.
3. **Regularization Matrix:** Constructing $\mathbf{R}$ involves $O(n_S \cdot m^2)$ for the variance term and $O(m^2)$ for the MMD term.
4. **SGD Iterations:** Each iteration processes $O(N)$ samples with $O(m)$ operations per sample, yielding $O(T \cdot N \cdot m)$ total complexity.

The overall time complexity is $O(T \cdot N \cdot m + m^3)$. Since $m \ll N$ (typically $m = O(\sqrt{N})$), this represents a substantial reduction from the $O(N^3)$ complexity of the standard QP solver. Table 1 summarizes the comparison between the two algorithms.

Algorithm 2: Fast TLQLDMR via Nyström Approximation

Input : Source data D_S , Target data D_T , Number of landmarks m , Quantile τ , Hyperparameters $\lambda_1, \lambda_2, C_S, C_T$, Learning rate η , Batch size B , Max iterations T .
Output: Optimal weights $\mathbf{w} \in \mathbb{R}^m$ and bias $b \in \mathbb{R}$.
// Step 1: Nyström Feature Mapping
1 Select m landmarks $Z = \{z_1, \dots, z_m\}$ from D_S via K-Means;
2 Compute landmark kernel matrix $\mathbf{W} \in \mathbb{R}^{m \times m}$;
3 Compute $\mathbf{W}^{-1/2}$ via eigendecomposition;
4 Construct explicit features $\psi(\mathbf{x})$ for all samples using Eq. (22);
// Step 2: Pre-compute Regularization Matrix
5 Compute feature matrices $\mathbf{\Psi}_S, \mathbf{\Psi}_T$ and mean vectors $\bar{\psi}_S, \bar{\psi}_T$;
6 Construct regularization matrix $\mathbf{R}$ using Eq. (23);
// Step 3: Stochastic Optimization
7 Initialize $\mathbf{w} \leftarrow \mathbf{0}, b \leftarrow 0$;
8 **for** $t = 1$ **to** T **do**
9 Shuffle combined dataset $D = D_S \cup D_T$;
10 **for** each mini-batch $B \subset D$ **do**
11 Compute predictions: $\hat{y}_k = \mathbf{w}^T \psi(\mathbf{x}_k) + b, \forall k \in B$;
12 Compute sub-gradients g_k using Eq. (24);
13 Apply sample weights: $\tilde{g}_k \leftarrow C_k \cdot g_k$;
14 Compute gradients $\nabla_{\mathbf{w}} J, \nabla_b J$ using Eq. (25);
15 Update $\mathbf{w}, b$ using Adam optimizer with learning rate η ;
16 **end**
17 **end**
18 **return** $\mathbf{w}, b$;

Table 1

Complexity Comparison between Standard QP and Scalable Algorithm

Component	Standard QP	Scalable (Ours)
Kernel/Feature Computation	$O(N^2 d)$	$O(N m d)$
Matrix Inversion	$O(N^3)$	$O(m^3)$
Optimization	$O(N^3)$	$O(T N m)$
Total	$O(N^3)$	$O(T N m + m^3)$

3.5.5. Theoretical Justification

The convergence of the proposed stochastic optimization algorithm is guaranteed under standard assumptions. Since the Pinball Loss is convex and Lipschitz continuous, and the regularization term $\frac{1}{2} \mathbf{w}^T \mathbf{R} \mathbf{w}$ is strongly convex (as $\mathbf{R}$ inherits positive definiteness from $\mathbf{\Omega}$ when $\lambda_1, \lambda_2 > 0$), the overall objective function is strongly convex. According to the convergence theory of SGD for strongly convex functions Bottou et al. (2018), the algorithm converges to the global optimum at a rate of $O(1/T)$.

Furthermore, the Nyström approximation error is bounded. Let $\tilde{k}(\mathbf{x}_i, \mathbf{x}_j) = \psi(\mathbf{x}_i)^T \psi(\mathbf{x}_j)$ denote the approximated kernel. Under mild conditions on the kernel eigenvalue decay (e.g., polynomial or exponential decay rate, which holds for commonly used kernels such as RBF and polynomial kernels), the approximation error satisfies:

$$\mathbb{E} [\|\mathbf{K} - \tilde{\mathbf{K}}\|_F] \leq O\left(\frac{N}{\sqrt{m}}\right) \quad (26)$$

This bound ensures that with a sufficient number of landmarks, the approximation quality is well-controlled, and the solution obtained by the scalable algorithm closely approximates that of the exact QP solver.

Table 2

Summary of wind farm sites used in the experiments.

Farm ID	Site Name	Nominal Capacity (MW)	Total Samples
1	Wind farm site 1	99	~70,000
2	Wind farm site 2	200	~70,000
3	Wind farm site 3	99	~70,000
4	Wind farm site 4	66	~70,000
5	Wind farm site 5	36	~70,000
6	Wind farm site 6	96	~70,000

4. Experiments

In this section, we conduct a comprehensive experimental evaluation of the proposed TLQLDMR model for wind power forecasting under extreme weather conditions. The experiments are organized into three progressive stages: (1) extreme weather sample screening and domain shift quantification across multiple wind farms (Experiment 1); (2) deterministic point prediction comparison on the wind farm exhibiting the most severe distribution shift (Experiment 2); and (3) probabilistic interval prediction evaluation under a 95% confidence level (Experiment 3). We first describe the dataset and experimental setup in Sections 4.1 and 4.4, then present the results and analysis in Sections 4.5–4.7.

4.1. Dataset Description

The experiments are conducted on a publicly available wind farm SCADA dataset published by the State Grid Renewable Energy Generation Dataset (State Grid Renewable Energy, 2022). The dataset contains supervisory control and data acquisition (SCADA) records from six onshore wind farms with nominal capacities ranging from 36 MW to 200 MW. Each record is sampled at 15-minute intervals and includes the following meteorological and operational variables:

- Wind speed at multiple heights: hub height, 50 m, 30 m, and 10 m (m/s);
- Wind direction at hub height (degrees);
- Air temperature (°C);
- Atmospheric pressure (hPa);
- Relative humidity (%);
- Active power output (MW).

Table 2 summarizes the basic characteristics of the six wind farms. Each farm contains approximately 70,000 time-stamped records spanning multiple years of continuous operation.

4.2. Feature Engineering and Data Preprocessing

The feature set comprises multi-height wind speeds, air temperature, atmospheric pressure, relative humidity, and cyclically encoded wind direction. Specifically, the raw wind direction angle θ is transformed into two orthogonal components $\sin(\theta)$ and $\cos(\theta)$ to eliminate the discontinuity at the $0^\circ/360^\circ$ boundary, which is a well-established practice in meteorological feature engineering (Wang et al., 2019).

A sliding window of length $W = 24$ (corresponding to a 6-hour lookback horizon at 15-minute resolution) is applied to construct supervised learning samples. For each window, the feature matrix $\mathbf{X}_t \in \mathbb{R}^{W \times d}$ is flattened into a vector $\mathbf{x}_t \in \mathbb{R}^{Wd}$, and the target variable y_t is the power output at the next time step. All features and the target variable are standardized to zero mean and unit variance using statistics computed from the training set. Missing values are imputed with the per-feature mean to ensure numerical stability for kernel-based models.

4.3. Extreme Weather Event Definition

A critical prerequisite for evaluating transfer learning under extreme conditions is the rigorous definition of what constitutes an “extreme weather event.” We adopt a rule-based identification scheme that combines statistical thresholds with physical domain knowledge. A sample is classified as belonging to the *target domain* (extreme weather) if it satisfies any of the following criteria:

1. **Statistical wind speed extreme:** Hub-height wind speed exceeds the 98th percentile of the farm’s historical distribution;
2. **Temperature extreme:** Air temperature falls below the 2nd percentile or exceeds the 98th percentile;
3. **Cut-out wind speed:** Hub-height wind speed exceeds 25 m/s, the typical cut-out threshold for modern wind turbines.

Notably, the power ramp criterion (rapid power fluctuations exceeding 5% of nominal capacity per 15-minute interval) is deliberately excluded from the extreme definition. This design choice avoids conflating control-induced power curtailment or dispatch adjustments with genuinely weather-driven anomalies. The union of the above three rules yields a target domain that is both physically interpretable and statistically sufficient for model training. All remaining samples constitute the *source domain* (normal weather conditions).

4.4. Experimental Setup

4.4.1. Evaluation Metrics

For deterministic point prediction (Experiment 2), we employ the following metrics. Let y_i and $\hat{y}_i$ denote the true and predicted power values, respectively, for $i = 1, \dots, n$:

- **Root Mean Squared Error (RMSE):** $\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
- **Mean Absolute Error (MAE):** $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$
- **Coefficient of Determination (R^2):** $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$

For probabilistic interval prediction (Experiment 3), let $[L_i, U_i]$ denote the predicted interval at confidence level $1 - \alpha$:

- **Prediction Interval Coverage Probability (PICP):** $\text{PICP} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(L_i \leq y_i \leq U_i)$
- **Mean Prediction Interval Width (MPIW):** $\text{MPIW} = \frac{1}{n} \sum_{i=1}^n (U_i - L_i)$
- **Prediction Interval Normalized Average Width (PINAW):** $\text{PINAW} = \frac{\text{MPIW}}{\max(y) - \min(y)}$
- **Coverage Width-based Criterion (CWC):** A composite metric that penalizes PINAW when PICP falls below the nominal coverage $1 - \alpha$:

$$\text{CWC} = \text{PINAW} \times (1 + \gamma(\text{PICP}) \cdot e^{-\eta(\text{PICP} - (1 - \alpha))}) \quad (27)$$

where $\gamma(\text{PICP}) = 1$ if $\text{PICP} < 1 - \alpha$ and 0 otherwise, and η is a penalty coefficient (set to 50 in our experiments);

- **Winkler Score:** An interval scoring rule that combines interval width with a penalty proportional to the magnitude of coverage violations:

$$\text{Winkler} = \frac{1}{n} \sum_{i=1}^n \begin{cases} (U_i - L_i) + \frac{2}{\alpha}(L_i - y_i), & y_i < L_i \\ (U_i - L_i), & L_i \leq y_i \leq U_i \\ (U_i - L_i) + \frac{2}{\alpha}(y_i - U_i), & y_i > U_i \end{cases} \quad (28)$$

PICP measures reliability (whether the interval achieves the target coverage), while MPIW and PINAW measure sharpness (how tight the interval is). An ideal model achieves $\text{PICP} \geq 1 - \alpha$ with the smallest possible PINAW.

Table 3

Domain shift metrics between normal and extreme weather domains across six wind farms. Farms are sorted by the composite Wasserstein distance in descending order.

Wind Farm	Extreme Samples	$W_1(\text{Wind})$	$W_1(\text{Power})$	KS(Wind)	KS(Power)
Farm 6 (96 MW)	4,144	4.060	21.057	0.389	0.344
Farm 3 (99 MW)	4,144	3.389	18.915	0.338	0.265
Farm 2 (200 MW)	4,211	3.467	16.166	0.333	0.142
Farm 5 (36 MW)	3,039	4.195	9.851	0.460	0.363
Farm 4 (66 MW)	4,046	4.235	7.759	0.347	0.201
Farm 1 (99 MW)	4,184	3.140	5.964	0.335	0.116

4.4.2. Domain Shift Metrics

To quantify the distribution discrepancy between normal and extreme domains (Experiment 1), we employ two complementary metrics computed on both wind speed and power dimensions:

- **Wasserstein-1 distance (W_1):** The earth mover's distance between the marginal distributions of the source and target domains, measuring the minimum "transportation cost" to transform one distribution into the other;
- **Kolmogorov-Smirnov statistic (KS):** The supremum of the absolute difference between the empirical cumulative distribution functions of the two domains.

4.4.3. Implementation Details

All experiments are implemented in Python using NumPy and scikit-learn. For SVM-based models, the training sample size is capped at 4,000 source domain samples and 1,200 target domain samples to maintain tractable computation times for kernel methods. Tree-based and ensemble models use the full training set without subsampling. The TLQLDMR model employs a Nyström approximation with $m = 1,200$ landmark points and is optimized via stochastic gradient descent with a learning rate of 0.01 for 64 epochs and a mini-batch size of 256. All random splits use a fixed seed for reproducibility.

4.5. Experiment 1: Extreme Sample Screening and Domain Shift Quantification

The first experiment identifies extreme-weather samples across all six wind farms using the criteria in Section 4.3, and then quantifies source-target distribution shift for each farm. Table 3 reports the domain-shift statistics sorted by the composite Wasserstein distance $W_1(\text{wind}) + W_1(\text{power})$. Preliminary sensitivity checks show that a stricter 99th-percentile wind-speed rule leaves too few target-domain samples for stable model development, whereas the adopted 98th-percentile rule preserves event extremity while maintaining statistical sufficiency.

Fig. 2 visualizes the distribution shift between normal and extreme weather domains for all six wind farms. Each subplot displays the kernel density estimates of wind speed (upper panel) and power output (lower panel) for both domains. The visual comparison confirms the quantitative findings in Table 3: Farm 6 exhibits the most pronounced separation between the two distributions, particularly in the power dimension, where the extreme domain shows a bimodal pattern with substantial mass near zero power (cut-out events) and high power (pre-cut-out high wind conditions).

Several observations emerge from Table 3 and Fig. 2. First, all six farms exhibit non-trivial distribution shifts between normal and extreme domains, with KS statistics ranging from 0.116 to 0.460, confirming that extreme weather events induce statistically significant distributional changes in both wind speed and power output. Second, the magnitude of the power distribution shift varies substantially across farms: Farm 6 exhibits a Wasserstein distance of 21.057 in the power dimension, approximately 3.5 times that of Farm 1 (5.964). This disparity reflects differences in turbine control strategies, terrain effects, and the local meteorological characteristics of each site.

Farm 6 (Nominal capacity 96 MW) ranks first in the composite domain shift metric ($W_1(\text{wind}) + W_1(\text{power}) = 25.117$), indicating the most pronounced distributional discrepancy between its normal and extreme operating regimes. Consequently, **Farm 6 is selected as the target wind farm for Experiments 2 and 3**, providing the most stringent test bed for evaluating the robustness of transfer learning methods under extreme weather conditions.

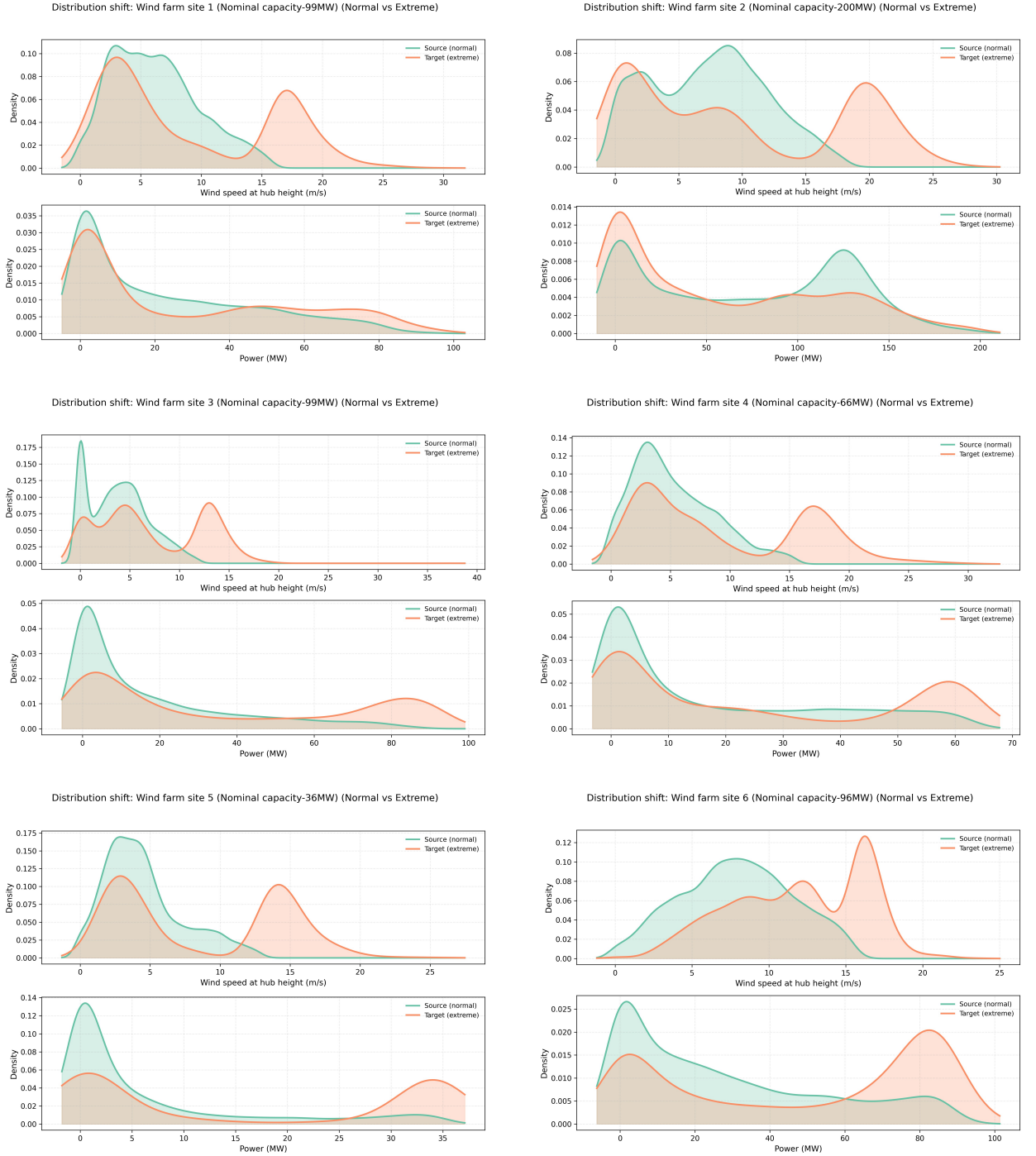


Fig. 2: Distribution shift visualization for all six wind farms. From left to right, top to bottom: Site 1 (99 MW), Site 2 (200 MW), Site 3 (99 MW), Site 4 (66 MW), Site 5 (36 MW), and Site 6 (96 MW). Each subplot shows kernel density estimates of wind speed (upper panel) and power output (lower panel) for normal (source) and extreme (target) domains. Site 6 exhibits the most severe domain shift in the power dimension.

4.6. Experiment 2: Deterministic Point Prediction under Extreme Weather

The second experiment evaluates deterministic point prediction on Farm 6 and compares TLQLDMR with six baselines under the same data split and preprocessing protocol.

The data partition for Farm 6 is as follows:

- **Source domain** (normal weather): 66,009 samples;
- **Target domain** (extreme weather): 4,143 samples, further split into:
 - Training set: 3,728 samples (90%);
 - Test set: 415 samples (10%).

The target domain is split via random shuffling with a fixed seed to ensure reproducibility. The 90%/10% target split is selected to maximize training signal under scarce extreme samples while retaining an independent hold-out subset for evaluation. All baseline models are trained exclusively on target-domain training data, whereas TLQLDMR additionally leverages source-domain samples through transfer learning.

We compare TLQLDMR against six state-of-the-art non-deep-learning models published in 2025, spanning SVM variants, tree-based ensembles, and random forest methods specifically designed for wind power forecasting:

1. **HHO-SVR** (Houssein et al., 2025): Support Vector Regression with hyperparameters optimized by the Harris Hawks Optimization metaheuristic;
2. **FLSVR** (Meena et al., 2025): A functional iterative method for solving the Lagrangian dual of SVR, offering improved convergence properties;
3. **ARA-SVR** (Duan et al., 2025): SVR with Associated Road Analysis for feature selection, originally proposed for short-term traffic flow prediction;
4. **KMeans-GBT** (Rizkallah, 2025): A two-stage model combining KMeans clustering with Gradient Boosted Trees to capture heterogeneous input–output relationships;
5. **RF-WPF** (Mustaffa & Sulaiman, 2025): A Random Forest model tailored for wind power forecasting with optimized tree depth and feature subsampling;
6. **BRF-WPF** (Chen & Shi, 2025): Broad Random Forest integrating Broad Learning System feature expansion with Random Forest regression for short-term wind power prediction.

All baseline models use hyperparameters consistent with their respective publications or default configurations, without extensive grid search, to ensure a fair comparison that reflects each method’s out-of-the-box performance.

The key hyperparameters of TLQLDMR are determined via a structured search on the validation set (20% held out from the target training data). The optimal configuration is: $\lambda_1 = 5 \times 10^{-4}$, $\lambda_2 = 1 \times 10^{-4}$, $C_S = 0.05$, $C_T = 150$, $\gamma_{\text{kernel}} = 0.002$, Nyström components $m = 1,200$, learning rate $\eta = 0.01$, epochs = 64, batch size = 256, and $\tau = 0.5$ (median regression for point prediction). The large ratio $C_T/C_S = 3,000$ reflects the asymmetric weighting strategy that assigns substantially higher importance to the scarce extreme weather samples.

Table 4 presents the test set performance of all models on the extreme weather domain of Farm 6.

Table 4 and Figs. 3–4 jointly demonstrate that TLQLDMR provides the most accurate and stable point forecasts on the Farm 6 extreme-weather test set. Several observations are noteworthy:

TLQLDMR improves on the strongest non-transfer baseline (HHO-SVR), raising R^2 from 0.9537 to 0.9553 and reducing RMSE from 7.603 to 7.476. Although the numerical margin is moderate, it is obtained under the most severe domain-shift setting with only 4,143 extreme-weather samples. The curve-level evidence in Figs. 3–4 is consistent with the aggregate metrics: TLQLDMR tracks abrupt ramps and low-power transitions more closely, while several baselines show larger local deviations around rapid fluctuations. Among baselines, HHO-SVR remains the most competitive kernel method, RF-WPF remains the second-best overall alternative, and BRF-WPF does not surpass RF-WPF under the present small-sample extreme-weather regime. The observed gain of TLQLDMR is consistent with the combined effect of source-domain transfer, asymmetric target weighting, and cyclic wind-direction encoding. Because many extreme-weather periods contain near-zero power outputs, MAPE is numerically unstable and therefore RMSE and R^2 are treated as the principal deterministic indicators.

To assess the robustness of the models against measurement noise, which is prevalent in SCADA systems during extreme weather, we evaluate the point prediction performance by injecting Gaussian noise into both the training and

Table 4

Deterministic point prediction results on the extreme weather test set of Farm 6 (96 MW). Best results are highlighted in bold.

Model	RMSE	MAE	R^2
HHO-SVR	7.603	5.530	0.9537
FLSVR	8.794	6.301	0.9381
ARA-SVR	9.246	5.954	0.9316
KMeans-GBT	9.620	6.695	0.9259
RF-WPF	7.967	5.344	0.9492
BRF-WPF	9.719	6.870	0.9244
TLQLDMR	7.476	5.348	0.9553

Table 5

Model performance (R^2) under varying Signal-to-Noise Ratios (SNR) on the extreme weather test set. Best results are highlighted in bold.

Model	SNR = 10 dB	SNR = 20 dB	SNR = 30 dB	SNR = 40 dB	SNR = 60 dB
HHO-SVR	0.9309	0.9539	0.9542	0.9538	0.9537
FLSVR	0.9253	0.9371	0.9380	0.9381	0.9381
ARA-SVR	0.9145	0.9303	0.9311	0.9316	0.9316
KMeans-GBT	0.9137	0.9247	0.9209	0.9230	0.9262
RF-WPF	0.9265	0.9438	0.9471	0.9485	0.9469
BRF-WPF	0.8939	0.9144	0.9177	0.9250	0.9253
TLQLDMR	0.9419	0.9547	0.9584	0.9560	0.9603

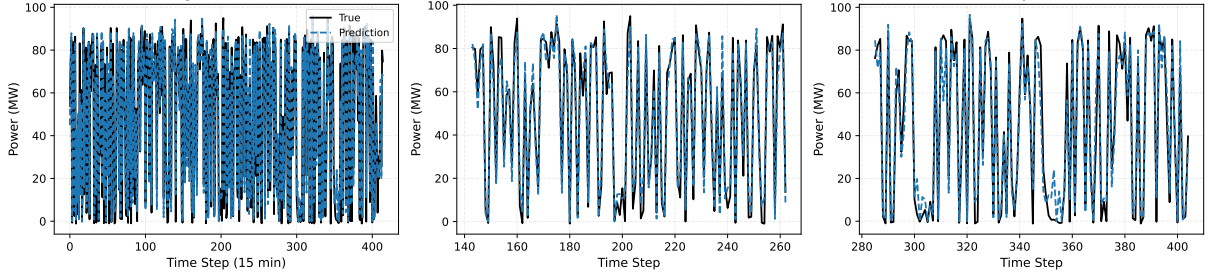
testing input features. We simulate varying Signal-to-Noise Ratios ($\text{SNR} \in \{60, 40, 30, 20, 10\}$ dB). For this analysis, TLQLDMR utilizes a higher-dimensional Nyström mapping ($m = 1,500$) and longer training duration (80 epochs) to accommodate the increased input variance.

The R^2 results under different noise levels are summarized in Table 5.

The observation confirms that while all models experience performance degradation as the noise level increases (SNR decreases), TLQLDMR consistently maintains the highest R^2 across all testing conditions. In particular, TLQLDMR remains above 0.94 even at 10 dB, while BRF-WPF exhibits the steepest degradation, indicating higher sensitivity to measurement perturbations. The transfer learning framework, reinforced by Large Margin Distribution regularization, effectively mitigates overfitting to noisy target samples and demonstrates the strongest noise resilience among the comparative models.

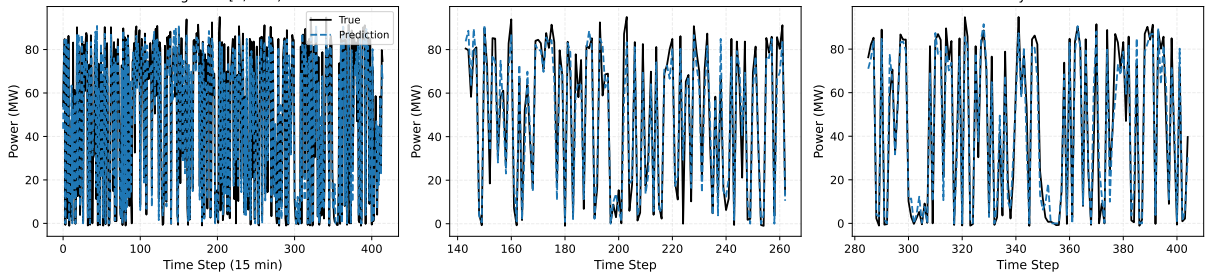
(a) HHO-SVR

HHO-SVR vs True (Wind farm site 6 (Nominal capacity-96MW))
Segment [0, 415]



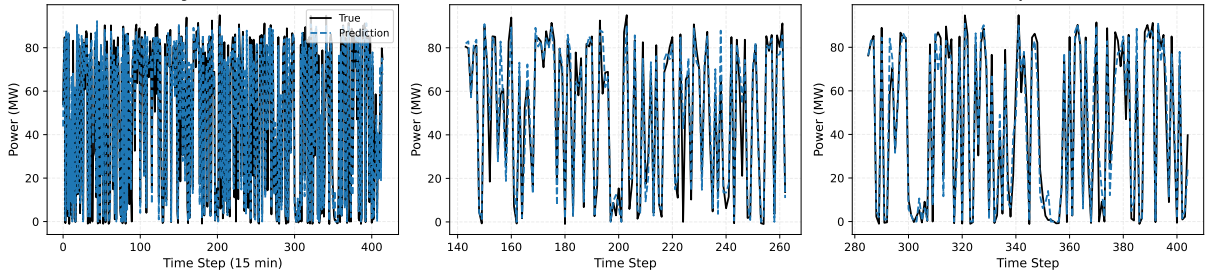
(b) FLSVR

FLSVR vs True (Wind farm site 6 (Nominal capacity-96MW))
Segment [0, 415]



(c) ARA-SVR

ARA-SVR vs True (Wind farm site 6 (Nominal capacity-96MW))
Segment [0, 415]



(d) KMeans-GBT

KMeans-GBT vs True (Wind farm site 6 (Nominal capacity-96MW))
Segment [0, 415]

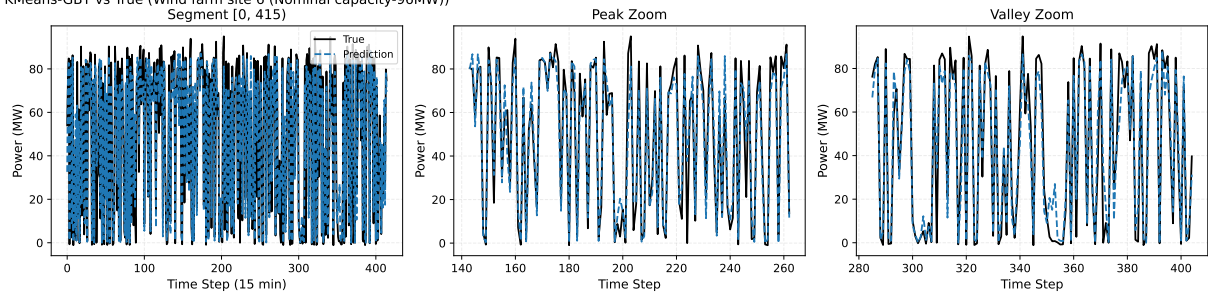
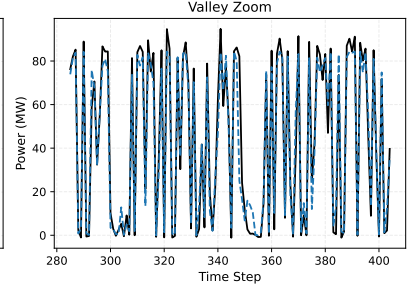
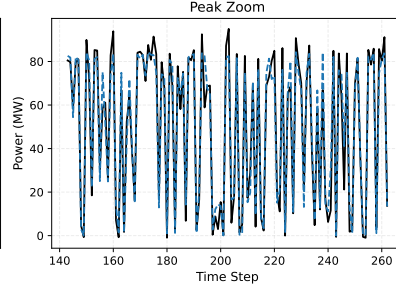
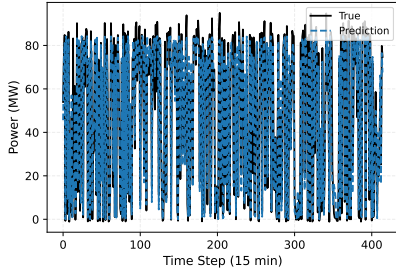
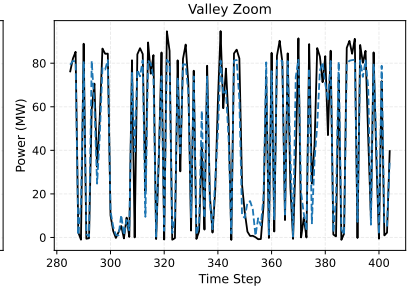
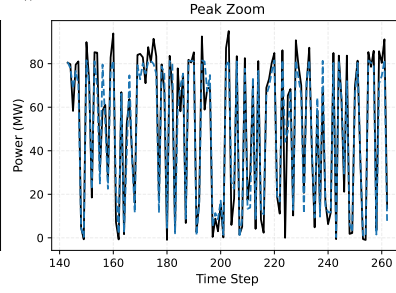
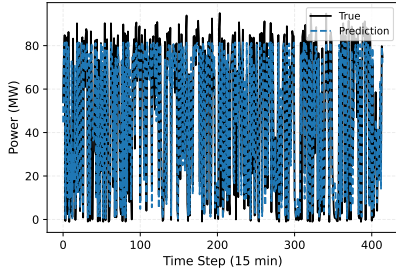


Fig. 3: Prediction-curve comparison on the Farm 6 extreme-weather test set (Part I). One row corresponds to one model, and each row contains the full segment plus peak/valley zoom views.

(e) RF-WPF

RF-WPF vs True (Wind farm site 6 (Nominal capacity-96MW))
Segment [0, 415)

(f) BRF-WPF

BRF-WPF vs True (Wind farm site 6 (Nominal capacity-96MW))
Segment [0, 415)

(g) TLQLDMR

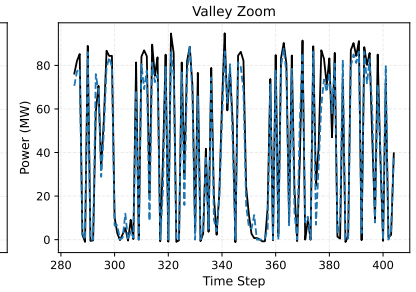
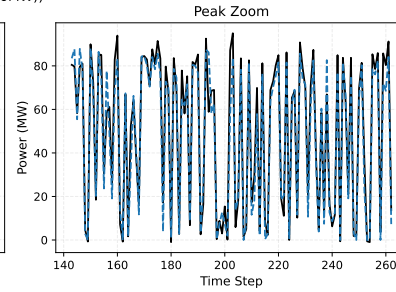
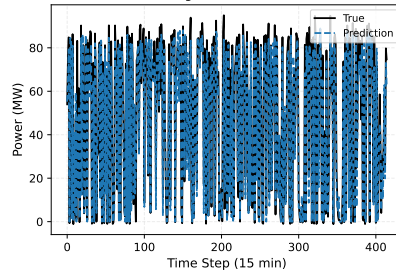
TL-QLDMR vs True (Wind farm site 6 (Nominal capacity-96MW))
Segment [0, 415)

Fig. 4: Prediction-curve comparison on the Farm 6 extreme-weather test set (Part II). One row corresponds to one model, and each row contains the full segment plus peak/valley zoom views.

4.7. Experiment 3: Probabilistic Interval Prediction under Extreme Weather

The third experiment extends evaluation from point prediction to probabilistic interval forecasting. The objective is to verify whether TLQLDMR can achieve reliable 95% prediction intervals while keeping interval width sufficiently sharp under extreme-weather volatility.

To support the calibration-based interval prediction framework, the target domain data is partitioned into four subsets:

- Training set: 2,983 samples (for model fitting);
- Validation set: 559 samples (for hyperparameter selection and width scaling);
- Calibration set: 745 samples (for Conformalized Quantile Regression adjustment);
- Test set: 415 samples (for final evaluation).

All models employ a two-stage calibration procedure based on Conformalized Quantile Regression (CQR):

1. **Conformal adjustment:** The model first predicts intervals on the calibration set. The conformal quantile $\hat{q}$ is computed as the $(1 - \alpha)$ -quantile of the nonconformity scores $\max(L_i - y_i, y_i - U_i)$, ensuring finite-sample coverage guarantees;
2. **Width scaling:** A scaling factor is selected on the validation set to optimize the trade-off between coverage and interval width, subject to the constraint that PICP approaches the target level $1 - \alpha + \delta$, where $\delta = 0.02$ is a safety margin.

All predicted intervals are clipped to the physical range $[0, P_{\text{rated}}]$, where P_{rated} is the nominal capacity of the wind farm.

The interval prediction baselines are organized into two categories:

SVM-based quantile regression variants:

1. **TSVQR** (Ye et al., 2023): Twin Support Vector Quantile Regression, which constructs separate hyperplanes for upper and lower quantiles;
2. **UQSVM-SSVQR** (Wang & Xu, 2024): Sparse Support Vector Quantile Regression with approximate kernel methods for scalability;
3. **NFS-SVQR** (Ye et al., 2025b): Nonlinear Feature Selection combined with SVQR, published in *Neural Networks* (2025);
4. **NuSVR-CI**: ν -Support Vector Regression with CQR-calibrated prediction intervals as a classical SVM baseline.

Recent wind power interval prediction models:

1. **HybridDL-Interval** (Mao et al., 2025): A hybrid deep learning framework for day-ahead wind power interval forecasting, combining temporal attention and quantile loss (approximate implementation);
2. **AMQRNN** (Ye et al., 2025a): A recurrent attention encoder-decoder network for multi-step interval wind power prediction, implemented using an attention-GRU network;
3. **QGBR** (Hu & Liu, 2025): Quantile Gradient Boosted Regression, representing the improved quantile ensemble learning framework for wind power prediction;
4. **CQR-MLP** (Hu et al., 2025): MLP quantile ensemble with dynamic conformalized quantile regression, acting as a simplified baseline for neural ensemble search methods.

For interval prediction, TLQLDMR trains two separate quantile models at τ_{low} and τ_{high} to construct the lower and upper bounds of the prediction interval. The quantile pair ($\tau_{\text{low}} = 0.2, \tau_{\text{high}} = 0.8$) is jointly selected with the hyperparameters on the validation set, using a coverage-priority criterion that first ensures PICP approaches the target coverage $0.95 + \delta$ and then minimizes PINAW via a learned interval scaling factor (width_scale). The resulting width scaling for TLQLDMR is 0.88, which yields reliable coverage while avoiding unnecessary interval inflation.

Table 6 presents the interval prediction results on the extreme weather test set of Farm 6 at the 95% confidence level ($\alpha = 0.05$).

The interval prediction results indicate a clear reliability-sharpness trade-off under extreme weather, and TLQLDMR offers the most favorable balance. TLQLDMR reaches the nominal 95% coverage target (PICP = 0.959)

Table 6

Interval prediction results on the extreme weather test set of Farm 6 at 95% confidence level ($\alpha = 0.05$). Baseline models are sorted by CWC in ascending order, while TLQLDMR is placed in the last row for direct comparison. Best results per column are highlighted in bold.

Model	PICP	MPIW	PINAW	CWC	Winkler
NuSVR-CI	0.843	19.970	0.208	0.285	54.603
HybridDL-Interval	0.954	31.645	0.329	0.329	60.224
TSVQR	0.889	24.592	0.256	0.342	48.178
QGBR	0.949	31.715	0.330	0.429	37.643
CQR-MLP	0.942	31.750	0.330	0.431	46.772
AMQRNN	0.940	32.432	0.337	0.441	66.874
NFS-SVQR	0.913	68.359	0.711	0.941	76.997
UQSVM-SSVQR	0.928	81.710	0.850	1.117	85.162
TLQLDMR	0.959	25.791	0.268	0.268	34.506

while obtaining the best composite quality (CWC = 0.268) and the best Winkler score (34.506), which shows that transfer learning with validation-based width scaling can provide both reliable and sharp intervals. By contrast, NuSVR-CI and TSVQR produce narrow intervals but severe under-coverage, whereas HybridDL-Interval, QGBR, CQR-MLP, and AMQRNN can approach acceptable coverage only with wider intervals. This comparison suggests that explicit asymmetric regularization combined with cross-domain adaptation is more effective for heavily skewed extreme-weather samples.

To assess interval robustness under sensor perturbation, Gaussian noise is injected only into test inputs at SNR $\in \{60, 40, 30\}$ dB, while calibration parameters remain fixed to preserve comparability across models. For TLQLDMR, a conservative scaling multiplier (1.15) is used to maintain coverage. Unlike the previous single-metric presentation, Table 7 reports the full set of interval metrics.

Across all noise levels, TLQLDMR maintains the best CWC while keeping PICP above 0.95. The CWC values remain tightly concentrated around 0.278, indicating stable interval quality under moderate measurement perturbation. In contrast, NuSVR-CI yields narrow intervals (small MPIW/PINAW) but exhibits severe under-coverage, and several deep/ensemble baselines require substantially wider intervals to approach reliable coverage. These results indicate that TLQLDMR preserves a stronger reliability-sharpness balance under measurement perturbations.

Table 7

Full interval metrics under injected test-input noise on the Farm 6 extreme-weather test set. For PICP, values near 0.95 are preferred; for MPIW, PINAW, CWC, and Winkler, lower values are preferred.

Model	SNR (dB)	PICP	MPIW	PINAW	CWC	Winkler
TSVQR	30	0.8892	24.570	0.2556	0.3422	48.306
	40	0.8892	24.588	0.2558	0.3424	48.140
	60	0.8892	24.592	0.2558	0.3425	48.169
UQSVM-SSVQR	30	0.9229	81.706	0.8499	1.1191	85.364
	40	0.9277	81.718	0.8501	1.1167	85.178
	60	0.9277	81.710	0.8500	1.1166	85.162
NFS-SVQR	30	0.9060	68.369	0.7112	0.9442	77.305
	40	0.9133	68.376	0.7113	0.9409	76.878
	60	0.9133	68.359	0.7111	0.9407	76.983
NuSVR-CI	30	0.8386	19.970	0.2077	0.2856	54.922
	40	0.8361	19.970	0.2077	0.2860	54.623
	60	0.8434	19.970	0.2077	0.2849	54.588
QGBR	30	0.9518	32.067	0.3336	0.3336	38.441
	40	0.9518	31.721	0.3300	0.3300	37.424
	60	0.9470	31.706	0.3298	0.4294	37.750
CQR-MLP	30	0.9325	32.548	0.3386	0.4438	48.828
	40	0.9325	32.550	0.3386	0.4438	48.780
	60	0.9325	32.554	0.3386	0.4438	48.743
HybridDL-Interval	30	0.9422	32.304	0.3360	0.4384	53.462
	40	0.9422	32.307	0.3361	0.4385	53.420
	60	0.9422	32.306	0.3361	0.4385	53.408
AMQRNN	30	0.9518	33.517	0.3487	0.3487	68.546
	40	0.9518	33.513	0.3486	0.3486	68.517
	60	0.9518	33.512	0.3486	0.3486	68.515
TLQLDMR	30	0.9542	26.709	0.2778	0.2778	36.307
	40	0.9566	26.713	0.2779	0.2779	36.034
	60	0.9566	26.715	0.2779	0.2779	36.083

4.8. Experiment 4: Ablation Study

To validate the contribution of the three core components (transfer learning, LDMR variance regularization, and MMD feature alignment), this ablation experiment evaluates structurally reduced variants on both point and interval tasks.

The full TLQLDMR framework is compared against three architectural reductions:

1. **w/o Transfer (Target-only QLDMR):** Trained exclusively on extreme weather target samples without source domain integration or MMD alignment.
2. **w/o LDMR (Variance Penalty):** The asymmetric variance regularizer is removed, deteriorating the framework into a generalized Transfer SVR/SVQR variant that lacks distributional robustness.
3. **w/o MMD (Feature Alignment):** The algorithm employs cross-domain training data but disables Maximum Mean Discrepancy alignment, relying purely on weighted objective loss.

Table 8 reports the ablation results on Farm 6. The full TLQLDMR consistently attains the best deterministic and probabilistic performance, indicating that all three components contribute non-trivially to final accuracy and calibration quality.

The ablation outcomes confirm that each component contributes materially to robustness. Removing transfer learning reduces both point accuracy (R^2 : 0.9553 $\rightarrow$ 0.9497) and interval quality (CWC: 0.2683 $\rightarrow$ 0.2797), indicating that target-only training is insufficient under strong domain shift. Removing LDMR regularization or MMD alignment causes clear interval-quality degradation (CWC rises to 0.3507 and 0.3528), even when point-metric changes are

Table 8

Ablation study evaluations for point and interval extreme-weather prediction. Best results are highlighted in bold.

Model Variants	Point Prediction		Interval Prediction (95% Level)			
	RMSE	R^2	PICP	MPIW	PINAW	CWC
w/o Transfer	7.930	0.9497	0.952	26.891	0.2797	0.2797
w/o LDMR	8.205	0.9461	0.949	25.925	0.2697	0.3507
w/o MMD	7.745	0.9520	0.947	26.049	0.2710	0.3528
TLQLDMR (Full)	7.476	0.9553	0.959	25.791	0.2683	0.2683

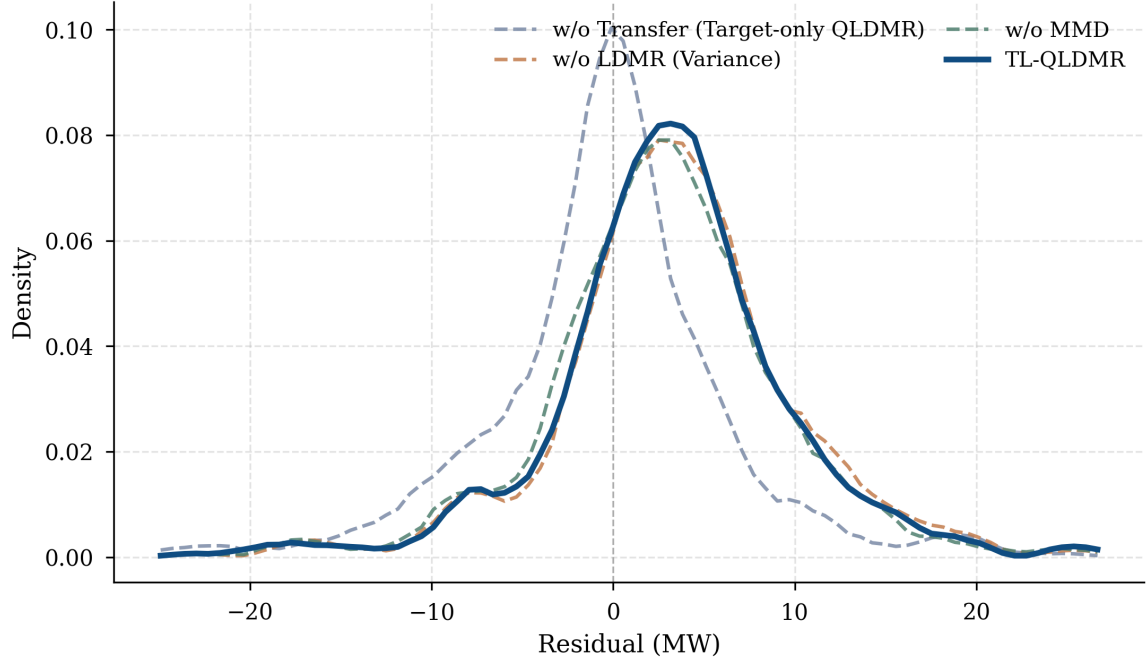


Fig. 5: Residual density comparison between TLQLDMR and ablated variants on the Farm 6 test set.

smaller, showing that residual-shape control and feature alignment are central to reliable uncertainty estimation. The residual-density comparison in Fig. 5 supports this conclusion by showing tighter central concentration and weaker heavy tails for the full TLQLDMR model; some ablated variants exhibit sharper central peaks but noticeably thicker tails, which corresponds to a higher probability of large-error events.

4.9. Experiment 5: Scalable Optimization Validation

Experiment 5 evaluates whether the proposed Nyström + SGD solver can preserve practical training efficiency as data scale increases, compared with three alternative optimization strategies implemented in Train_TL_QLDMR: (i) full dual QP via CVXOPT, (ii) gradient-based QP via PyTorch (torch_gd), and (iii) kernel-slice mini-batch SGD (batch_gsd).

The combined training size N is progressively increased from 100 to 2,000. For each configuration, wall-clock training time (seconds) is recorded. Table 9 summarizes the results.

Both CVXOPT and torch_gd become infeasible beyond $N = 800$ due to cubically growing memory and computation demands. The kernel-slice batch_gsd avoids full matrix inversion but still requires $O(N^2)$ kernel evaluations per batch, yielding runtimes of 3–6 seconds. In contrast, the proposed Nyström + SGD solver maintains near-linear scaling and completes training for $N = 2,000$ in only 1.67 seconds, confirming the practical viability of the approximate primal formulation for large-scale wind farm datasets.

Table 9

Training time (seconds) of four solvers over expanding dataset scales. “—” denotes infeasible runs due to excessive resource consumption.

N	CVXOPT	torch_gd	batch_sgd	Nyström + SGD (Ours)
100	0.494	2.254	2.972	0.024
200	0.662	2.398	2.300	0.554
400	0.884	2.508	4.374	0.696
600	2.728	2.668	4.874	0.824
800	4.949	2.375	3.890	0.924
1,200	—	—	6.049	1.114
1,600	—	—	6.123	1.465
2,000	—	—	—	1.670

4.10. Discussion

The extended experimental evaluations yield the following conclusive insights:

Transfer learning improves deterministic forecasting under severe shift. Experiment 2 shows consistent gains in RMSE and R^2 , and curve-level evidence confirms better tracking during abrupt peak-valley transitions.

Interval forecasting achieves a better reliability-sharpness balance. In Experiment 3, TLQLDMR achieves the highest coverage reliability among the compared methods while also obtaining the best composite penalty (CWC), and this advantage persists under injected measurement noise.

Ablation confirms the necessity of each component. Removing transfer learning reduces both point and interval performance under severe domain shift; removing asymmetric LDMR variance regularization degrades distributional robustness and interval calibration; removing MMD alignment weakens feature-space adaptation. Residual-density plots provide complementary evidence for this degradation.

The scalable optimizer materially reduces computational burden. Experiment 5 verifies that the Nyström + SGD formulation remains practical in regimes where full QP optimization fails due to computational constraints.

Limitations and future work. The present evaluation is based on onshore wind farms and fixed feature engineering pipelines; extension to offshore assets, dynamic re-calibration strategies, and hybrid deep-kernel transfer formulations remains a relevant direction.

5. Conclusion

This paper presents TLQLDMR for wind power forecasting under extreme weather, where target-domain samples are limited and distribution shift is pronounced. The method combines asymmetric variance regularization, MMD-based feature alignment, and weighted pinball loss within a unified transfer-learning quantile framework, with theoretical guarantees on positive definiteness and convergence.

Experimental results show that TLQLDMR delivers consistent gains in both deterministic and interval forecasting on the most challenging target farm. For point prediction, it achieves the best RMSE (7.476) and R^2 (0.9553). For 95% interval prediction, it attains PICP = 0.959 together with the best CWC (0.268), indicating a favorable reliability-sharpness trade-off. Under injected noise, TLQLDMR remains the strongest model across reported point and interval metrics. In addition, the Nyström + SGD solver provides practical scalability, reaching 1.67 seconds at $N = 2,000$ while full QP alternatives become infeasible beyond $N = 800$.

Future work will focus on extending the framework to multi-site spatiotemporal coupling, adaptive online recalibration under non-stationary weather, and hybrid kernel-deep transfer formulations.

CRedit authorship contribution statement

***: Methodology, Software, Writing - original draft. ***: Visualization, Data curation. ***: Formal analysis, Software. ***: Writing – review & editing. ***: Funding acquisition, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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